Yes, **masked image modeling is effective** for **remote sensing foundation models**, but task-specific adaptations usually work best.

## 1. Introduction

Masked image modeling is now one of the dominant pretraining strategies for remote sensing foundation models, and the evidence across this corpus supports it as an effective approach for learning transferable representations from large unlabeled geospatial imagery collections. Remote-sensing-specific MIM models repeatedly report gains across classification, segmentation, detection, change detection, retrieval, and multimodal transfer, often outperforming ImageNet pretraining or prior self-supervised baselines when the masking strategy and reconstruction target are adapted to remote-sensing structure (Sun et al., 2023; Wang et al., 2024; Dong et al., 2024; Zhang et al., 2023; Fuller et al., 2023; Fuller et al., 2022).

The core reason is practical as much as methodological: remote sensing has vast unlabeled archives, substantial domain gap from natural images, and many downstream tasks that benefit from pixel-level or object-level features rather than only image-level semantics (Sun et al., 2023; Lu et al., 2025; Lu et al., 2024; Zhang et al., 2023; Spradlin et al., 2024). At the same time, the evidence is not that vanilla MAE is universally optimal. Several studies show that random high-ratio masking can erase small objects, overemphasize low-level texture, and underperform on some low-label or pixel-level tasks unless the masking, decoder, or targets are redesigned for spectral, spatial, temporal, or multimodal remote-sensing data ( Liu et al., 2023; Zhang et al., 2023; Liu et al., 2026; Xie et al., 2024; Xu et al., 2025).

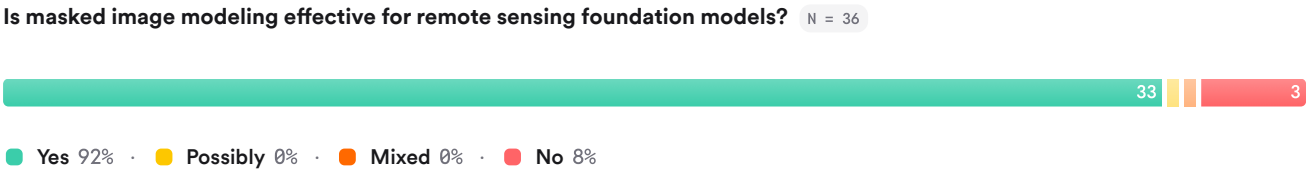



FIGURE 1 Consensus on MIM effectiveness in remote sensing

The meter points to a strong but not absolute consensus. The broad agreement is that MIM works well as a foundation-model pretraining objective in remote sensing, while the main disagreement concerns where plain random-masking MAE breaks down and how much remote-sensing-specific structure must be injected to recover strong transfer (Sun et al., 2023; Lu et al., 2025; Choudhury et al., 2025; Liu et al., 2026).

## 2. Methods

This deep search ran over more than 170 million research papers indexed in Consensus, spanning Semantic Scholar, PubMed, and other scholarly sources. The workflow identified 149 unique candidate papers after broad search expansion and citation crawling, then screened and ranked them for direct relevance to masked image modeling, masked autoencoders, and remote sensing foundation models. Fifty papers were included in the final synthesis, with emphasis on surveys, foundation-model papers, benchmark-driven empirical studies, and papers explicitly testing limitations or alternatives.

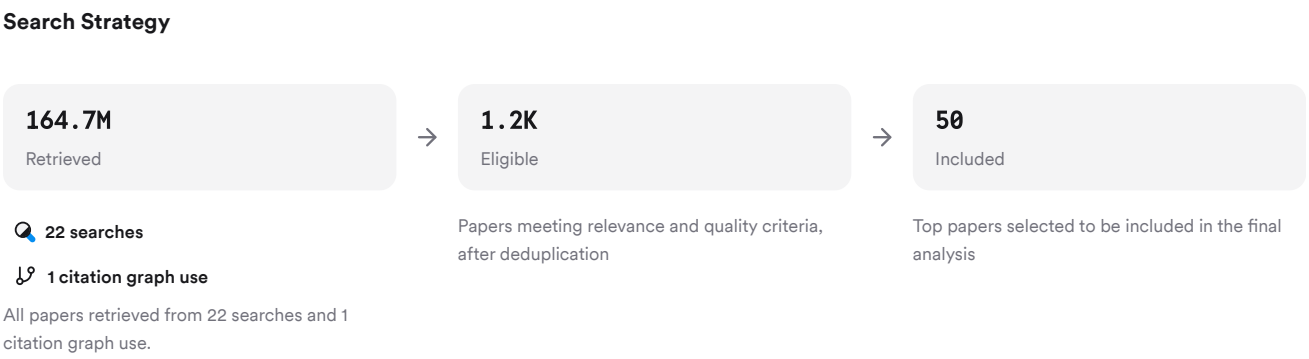



FIGURE 2 Deep search screening and paper inclusion flow

Searches combined MIM, MAE, self-supervised pretraining, multimodal geospatial models, transfer performance, and failure-mode keywords across remote sensing tasks.

## 3. Results

### 3.1 Key Papers

A small set of papers anchors the literature because they either established MIM-based remote sensing foundation modeling at scale or synthesized its performance across many tasks. RingMo is the clearest early landmark for in-domain MIM pretraining, CtxMIM is a strong demonstration that remote-sensing-aware masking improves transfer, and later surveys consolidate the view that MAE-style pretraining is central to the field (Sun et al., 2023; ■ Zhang et al., 2023; ■ Lu et al., 2025).

| Paper | Summary |
|---|---|
| (Sun et al., 2023) | Establishes RS MIM foundation modeling at scale (Sun et al., 2023) |
| ( ■ Lin et al., 2023) | Shows RS-aware MIM beats strong baselines ( ■ Zhang et al., 2023) |
| ( ■ Spradlin et al., 2024) | Survey finds MAE pretraining boosts robustness ( ■ Lu et al., 2025) |

FIGURE 3 Key anchor papers in the included corpus

### 3.2 Performance Across Tasks

The descriptive pattern is consistent: MIM-based pretraining improves representation quality across a wide task range when the downstream task depends on spatial context, local structure, or multispectral relations. RingMo reached state of the art on eight datasets across four tasks, GeoSense SMLFR reached state of the art across five datasets and three tasks, and SelectiveMAE improved classification, detection, and segmentation while doubling training efficiency (Sun et al., 2023; ■ Dong et al., 2024; ■ Wang et al., 2024).

| Dimension | Evidence Across Papers | Outcome | Citations |
|---|---|---|---|
| Classification | Scene, land-cover, hyperspectral | Usually improved transfer | ( ■ Gao et al., 2022; ■ Scheibenreif et al., 2023; ■ Lin et al., 2023) |
| Segmentation | Semantic and building tasks | Often improved, but task-sensitive | ( ■ Zhang et al., 2023; Cai et al., 2024; ■ Rodríguez et al., 2026) |

| Dimension | Evidence Across Papers | Outcome | Citations |
|---|---|---|---|
| Detection | Object-centric and aerial detection | Benefits from RS-aware masking | ( Zhang et al., 2023) |
| Change Detection | Multitemporal transfer | Strong gains with temporal MIM variants | (Zhang et al., 2023; Wan et al., 2025) |
| Multimodal Transfer | Optical-SAR or multisensor fusion | Better robustness and generalization | ( Fuller et al., 2023; Linial et al., 2025; Du et al., 2024) |

FIGURE 4 Task areas where MIM shows benefits

The largest gains often appear when MIM is extended beyond plain RGB patch reconstruction. Hyperspectral masked pretraining improved accuracy by 5% on EnMAP and 15% on Houston2018, with 7–8% gains in low-label settings, while FG-MAE improved SAR transfer by up to 5% over standard MAE ( Scheibenreif et al., 2023; Wang et al., 2023).

### 3.3 Why Variants Beat Vanilla MAE

The literature repeatedly argues that remote sensing requires masking and targets that reflect sensor physics, spectral structure, and small-object density rather than random token removal alone. CtxMIM adds context consistency to address dense scenes, Object-Centric MIM guides masking toward attended objects for detection, SS-MAE masks both spatial patches and spectral channels, and SIGMAE uses spectral indices to steer dynamic masking toward informative regions ( Zhang et al., 2023; Lin et al., 2023; Zhang et al., 2026).

A second theme is efficiency. Remote-sensing images often contain large redundant backgrounds, so reconstructing every masked token can waste computation. SelectiveMAE addresses this by reconstructing semantically rich patches and reports 2.2–2.7× efficiency gains, while CSMoE reports more than 2× computational efficiency with competitive or improved representational performance on top of a cross-sensor MAE design ( Wang et al., 2024; Hackel et al., 2025).

A third theme is modality expansion. MIM works not only for optical imagery but also for multispectral, hyperspectral, SAR, multimodal optical-SAR fusion, season-aware modeling, and coarse all-sky Earth observation, with strong results reported by S2MAE, CROMA, SeaMo, SatVision-TOA, FLORO, and mixed-modality MAE studies ( Li et al., 2024; Fuller et al., 2023; Li et al., 2024; Spradlin et al., 2024; Rodríguez et al., 2026; Linial et al., 2025).

### 3.4 Limits and Counterevidence

The main limitation is not whether MIM can work, but where standard MIM is poorly aligned with the downstream task. RemoteCLIP argues that MIM tends to learn low-level features with weaker semantics and is less suitable for zero-shot or few-shot semantic recognition than vision-language pretraining, while pixel-level classification work finds that foundation models are not always better than traditional ML when texture cues are weak ( Liu et al., 2023; Xie et al., 2024).

Small objects and low-label dense prediction are the most frequent failure cases. Random masking can entirely hide critical small targets, impairing convergence and local-feature learning, which motivated object-centric, edge-guided, gradient-guided, and complementary masking strategies ( Zhang et al., 2023; Liu et al., 2026; Zhang et al., 2023; Xu et al., 2025; Wang et al., 2022). Some image-to-image tasks also remain challenging: one review notes that MAE-based remote-sensing foundation models can trail a basic UNet on some such tasks, and reconstruction-based objectives do not always benefit from model-slimming protocols ( Duc & Fukui, 2025; Hackel et al., 2026).

### Results Timeline

Remote-sensing MIM has progressed from early proof-of-concept to scaled, modality-aware foundation modeling.

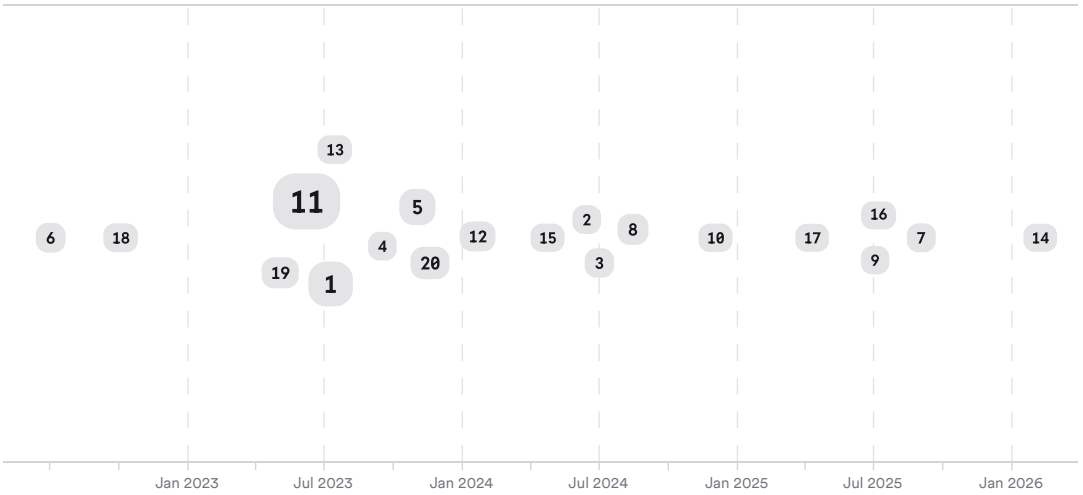



FIGURE 5 Research timeline with larger markers for citations

The timeline shows a fast arc. Early work established that masked pretraining can transfer to remote-sensing classification, then the field shifted toward task-aware and modality-aware designs, and recent papers focus on scaling, efficiency, multimodality, and failure-mode correction ( ■ Gao et al., 2022; Sun et al., 2023; ■ Wang et al., 2024; ■ Li et al., 2024).

**Top Contributors**

| Type | Name | Papers |
|---|---|---|
| Author | Xuyang Li | [7b088ef9ea33547286dee86d23231b94] [e63a3aeb4b745cb7aaf29768e3655535] |
| Author | Xiao Xiang Zhu | [bf23d9a15f1057d783aa64199d6f338b] [3de35990fd64560a8ee97ce33705c780] |
| Author | A. Fuller | [dac82254f6d25bfa85749bb85f32d1fe] [256a74a29da7502dafa88104480ab48a] |
| Journal | *IEEE Transactions on Geoscience and Remote Sensing* | [a564b730d18a5621bea626d76d33de76] [4467cbd5977d58098ce6fe76e7413347] [adca17c259785a9291e757d3b2173961] |
| Journal | *ArXiv* | [dac82254f6d25bfa85749bb85f32d1fe] [a10acdd480b95f0d8d6c0c2cd577bc4c] [22ae5bb25b115f3685bd1da53a538cc3] |
| Journal | *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* | [bf23d9a15f1057d783aa64199d6f338b] [48b4623d17f0582baddf130cf2a95030] |

FIGURE 6 Authors and journals most frequent in included papers

## 4. Discussion

The strongest claim supported by this corpus is that MIM is an effective pretraining family for remote sensing foundation models when judged by downstream transfer across many tasks and modalities. That conclusion rests on multiple large-scale foundation-model papers, broad surveys, and repeated reports of superiority over supervised or weaker self-supervised baselines in classification, segmentation, detection, retrieval, and change detection (Sun et al., 2023; Lu et al., 2025; Lu et al., 2024; Choudhury et al., 2025; Fuller et al., 2023).

The evidence is weaker for stronger claims such as “plain MAE is enough” or “MIM is best for every remote-sensing task.” Many positive results come from modified MIM designs that inject spatial-spectral priors, object awareness, multimodal structure, or task-specific masking. That pattern suggests the field’s success comes less from generic reconstruction alone than from using MIM as a flexible scaffold for remote-sensing inductive bias ( Dong et al., 2024; Wang et al., 2024; Lin et al., 2023; Linial et al., 2025; Rodríguez et al., 2026).

The main validity concern is task dependence. Low-level feature learning is useful for dense prediction, but it can be insufficient for semantic recognition, zero-shot use, or tasks where texture is not informative. The field increasingly answers this by hybridizing MIM with contrastive, language, temporal, cross-modal, or knowledge-guided objectives rather than abandoning MIM outright ( Liu et al., 2023; Muhtar et al., 2023; Fuller et al., 2023; Liu et al., 2023; Ravirathinam et al., 2024).

| Claim | Evidence Strength | Reasoning | Papers |
|---|---|---|---|
| MIM is effective overall for RS foundation-model pretraining | Strong | Multiple foundation-model papers, surveys, and cross-task benchmarks converge on positive transfer. | RingMo, CtxMIM, surveys, SatViT, CROMA (Sun et al., 2023; Zhang et al., 2023; Lu et al., 2025; Fuller et al., 2022; Fuller et al., 2023) |
| RS-specific MIM variants usually outperform vanilla masking | Strong | Many papers improve results by changing masking, targets, or modality handling. | OCMIM, SS-MAE, FG-MAE, SIGMAE, GA-MAE ( Zhang et al., 2023; Lin et al., 2023; Wang et al., 2023; Zhang et al., 2026; Xu et al., 2025) |
| MIM is especially useful for dense, spectral, or multimodal tasks | Moderate | Strong repeated task evidence, but not all downstream settings are equally favorable. | S2MAE, SeaMo, CROMA, mixed-modality MAE, change detection ( Li et al., 2024; Fuller et al., 2023; Linial et al., 2025; Wan et al., 2025) |
| Vanilla random-masking MAE is sufficient for all RS tasks | Weak | Directly contradicted by many studies reporting small-object loss, weak semantics, and task mismatch. | RemoteCLIP critique, low-label segmentation, pixel-level suitability studies ( Liu et al., 2023; Liu et al., 2026; Xie et al., 2024) |
| MIM alone is the best route to semantic or zero-shot RS understanding | Weak | Evidence favors complementary objectives for higher-level semantics. | RemoteCLIP, CMID, CROMA ( Liu et al., 2023; Muhtar et al., 2023; Fuller et al., 2023) |

FIGURE 7 Key claims and supporting evidence in corpus

## 5. Conclusion

Across these 50 papers, masked image modeling emerges as an effective and widely validated pretraining objective for remote sensing foundation models. Its strongest support is for transfer learning from large unlabeled archives into detection, segmentation, classification, change detection, and multimodal Earth-observation tasks, especially when the model respects remote-sensing-specific spectral, spatial, temporal, or sensor structure (Sun et al., 2023; Wang et al., 2024; Li et al., 2024; Spradlin et al., 2024).

The key nuance is that effectiveness is conditional. The corpus does not support a blanket claim that off-the-shelf MAE is universally optimal; instead, the best-performing models usually modify masking, reconstruction, or objectives to protect small objects, inject semantics, or align with downstream task structure ( Zhang et al., 2023; Liu et al., 2026; Zhang et al., 2026; Linial et al., 2025).

### Research Gaps

The most persistent gaps are low-label dense prediction, semantic richness beyond reconstruction, cross-sensor robustness, and clearer understanding of when MIM should yield to alternative pretraining objectives.

| Topic/Outcome | Task Breadth | Low-Label Tests | Multisensor Coverage | Mechanistic Analysis |
|---|---|---|---|---|
| Classification transfer | 10 | 6 | 5 | 3 |
| Segmentation transfer | 8 | 4 | 3 | 3 |
| Object detection and small objects | 6 | 2 | 2 | 3 |
| Cross-sensor and multimodal transfer | 6 | 2 | 7 | 2 |
| Zero-shot and semantic understanding | 3 | 1 | 2 | 2 |

FIGURE 8 Research gaps across remote sensing MIM topics

### Open Research Questions

The next phase of the field is likely to focus less on proving that MIM works and more on determining where and how it should be modified or combined.

| Question | Why |
|---|---|
| **Which remote-sensing tasks truly benefit from reconstruction-based pretraining over contrastive or vision-language objectives?** | Current evidence shows strong transfer overall, but semantic and texture-light tasks remain inconsistent and may require different objectives. |
| **What masking strategy best preserves small objects and local structure without sacrificing scalability?** | Random masking often hides critical targets, and many successful variants differ mainly in how they steer masking. |
| **How can MIM-based models gain stronger zero-shot semantics while keeping dense-prediction strength?** | MIM learns useful local features, but several papers argue that higher-level semantics remain a relative weakness. |

Masked image modeling is effective for remote sensing foundation models, but the evidence favors **remote-sensing-aware MIM** over vanilla MAE as the durable conclusion.

*These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.*

## References


Cai, M., Chen, H., Zhang, T., Yin, Z., & Chen, L. (2024). Consistency Regularization Based on Masked Image Modeling for Semisupervised Remote Sensing Semantic Segmentation. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 17*, 17442-17460. https://doi.org/10.1109/jstars.2024.3435509

Choudhury, S., Vasim, A., Schmitt, M., & Banerjee, B. (2025). MIMRS: A Survey on Masked Image Modeling in Remote Sensing. *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*, 2309-2314. https://doi.org/10.1109/igarss55030.2025.11243448

Dong, Z., Gu, Y., & Liu, T. (2024). Generative ConvNet Foundation Model With Sparse Modeling and Low-Frequency Reconstruction for Remote Sensing Image Interpretation. *IEEE Transactions on Geoscience and Remote Sensing, 62*, 1-16. https://doi.org/10.1109/tgrs.2023.3348479

Du, D., Liu, T., & Gu, Y. (2024). A Multimodal Unified Representation Learning Framework With Masked Image Modeling for Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing, 62*, 1-16. https://doi.org/10.1109/tgrs.2024.3494244

Duc, C., & Fukui, H. (2025). SatMamba: Development of Foundation Models for Remote Sensing Imagery Using State Space Models. *IOP Conference Series: Earth and Environmental Science, 1633*. https://doi.org/10.1088/1755-1315/1633/1/012014

Fuller, A., Millard, K., & Green, J. R. (2023). CROMA: Remote Sensing Representations with Contrastive Radar-Optical Masked Autoencoders. *ArXiv, abs/2311.00566*. https://doi.org/10.48550/arxiv.2311.00566

Fuller, A., Millard, K., & Green, J. R. (2022). SatViT: Pretraining Transformers for Earth Observation. *IEEE Geoscience and Remote Sensing Letters, 19*, 1-5. https://doi.org/10.1109/lgrs.2022.3201489

Gao, Y., Sun, X., & Liu, C. (2022). A General Self-Supervised Framework for Remote Sensing Image Classification. *Remote. Sens., 14*, 4824. https://doi.org/10.3390/rs14194824

Hackel, L., Burgert, T., & Demir, B. (2025). CSMoE: An Efficient Remote Sensing Foundation Model with Soft Mixture-of-Experts. *ArXiv, abs/2509.14104*. https://doi.org/10.48550/arxiv.2509.14104

Hackel, L., Burgert, T., & Demir, B. (2026). How Much of a Model Do We Need? Redundancy and Slimmability in Remote Sensing Foundation Models. *ArXiv, abs/2601.22841*. https://doi.org/10.48550/arxiv.2601.22841

Li, X., Hong, D., & Chanussot, J. (2024). S2MAE: A Spatial-Spectral Pretraining Foundation Model for Spectral Remote Sensing Data. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 27696-27705. https://doi.org/10.1109/cvpr52733.2024.02616

Li, X., Li, C., Vivone, G., & Hong, D. (2024). SeaMo: A season-aware multimodal foundation model for remote sensing. *Inf. Fusion, 125*, 103334. https://doi.org/10.1016/j.inffus.2025.103334

Lin, J., Gao, F., Shi, X., Dong, J., & Du, Q. (2023). SS-MAE: Spatial–Spectral Masked Autoencoder for Multisource Remote Sensing Image Classification. *IEEE Transactions on Geoscience and Remote Sensing, 61*, 1-14. https://doi.org/10.1109/tgrs.2023.3331717

Linial, O., Leifman, G., Blau, Y., Sherman, N., Gigi, Y., Sirko, W., Beryozkin, G., & Research, G. (2025). Enhancing Remote Sensing Representations Through Mixed-Modality Masked Autoencoding. *2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 470-479. https://doi.org/10.1109/wacvw65960.2025.00058

Liu, F., Chen, D., Guan, Z.-R., Zhou, X., Zhu, J., & Zhou, J. (2023). RemoteCLIP: A Vision Language Foundation Model for Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing, 62*, 1-16. https://doi.org/10.1109/tgrs.2024.3390838

Liu, P., Zhu, H., Mi, X., Yang, J., Meng, Y., Zhao, H., & Gu, X. (2026). A Systematic Study on Pretraining Strategies for Low-Label Remote Sensing Image Semantic Segmentation. *Sensors (Basel, Switzerland), 26*. https://doi.org/10.3390/s26041385

Lu, S., Guo, J., Zimmer-Dauphinee, J., Nieusma, J. M., Wang, X., VanValkenburgh, P., Wernke, S. A., & Huo, Y. (2025). Vision Foundation Models in Remote Sensing: A survey. *IEEE Geoscience and Remote Sensing Magazine, 13*, 190-215. https://doi.org/10.1109/mgrs.2025.3541952

Lu, S., Guo, J., Zimmer-Dauphinee, J., Nieusma, J. M., Wang, X., VanValkenburgh, P., Wernke, S. A., & Huo, Y. (2024). Vision Foundation Models in Remote Sensing: A Survey. https://doi.org/10.48550/arxiv.2408.03464

Muhtar, D., Zhang, X.-L., Xiao, P., Li, Z., & Gu, F.-X. (2023). CMID: A Unified Self-Supervised Learning Framework for Remote Sensing Image Understanding. *IEEE Transactions on Geoscience and Remote Sensing, 61*, 1-17. https://doi.org/10.1109/tgrs.2023.3268232

Ravirathinam, P., Khandelwal, A., Ghosh, R., & Kumar, V. (2024). Towards Knowledge Guided Pretraining Approaches for Multimodal Foundation Models: Applications in Remote Sensing. *ACM Transactions on Spatial Algorithms and Systems*. https://doi.org/10.1145/3806229

Rodríguez, J. L., Morales, V. A., Alwahas, A., Lara, M. E., Thoker, F. M., Johansen, K., Ghanem, B., Maestre, F. T., & McCabe, M. F. (2026). FLORO: A Multimodal Geospatial Foundation Model for Ecological Remote Sensing Across Sensors and Scales. https://doi.org/10.48550/arxiv.2605.28174

Scheibenreif, L., Mommert, M., & Borth, D. (2023). Masked Vision Transformers for Hyperspectral Image Classification. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2166-2176. https://doi.org/10.1109/cvprw59228.2023.00210

Spradlin, C., Caraballo-Vega, J., Li, J., Carroll, M., Gong, J., & Montesano, P. (2024). SatVision-TOA: A Geospatial Foundation Model for Coarse-Resolution All-Sky Remote Sensing Imagery. *ArXiv, abs/2411.17000*. https://doi.org/10.48550/arxiv.2411.17000

Sun, X., Wang, P., Lu, W., Zhu, Z., Lu, X., He, Q., Li, J., Rong, X., Yang, Z., Chang, H., He, Q., Yang, G., Wang, R., Lu, J., & Fu, K. (2023). RingMo: A Remote Sensing Foundation Model With Masked Image Modeling. *IEEE Transactions on Geoscience and Remote Sensing, 61*, 1-22. https://doi.org/10.1109/tgrs.2022.3194732

Wan, L., Xiang, Y., Kang, W., & L. (2025). A Self-Supervised Learning Pretraining Framework for Remote Sensing Image Change Detection. *IEEE Transactions on Geoscience and Remote Sensing, 63*, 1-16. https://doi.org/10.1109/tgrs.2025.3579416

Wang, F., Wang, H., Wang, D., Guo, Z., Zhong, Z., Lan, L., Yang, W., & Zhang, J. (2024). Harnessing Massive Satellite Imagery with Efficient Masked Image Modeling. *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, 6935-6947. https://doi.org/10.1109/iccv51701.2025.00652

Wang, Y., Hern'andez, H. H., Albrecht, C., & Zhu, X. X. (2023). Feature Guided Masked Autoencoder for Self-Supervised Learning in Remote Sensing. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 18*, 321-336. https://doi.org/10.1109/jstars.2024.3493237

Wang, F., Wang, H., Wang, D., Guo, Z., Zhong, Z., Lan, L., Zhang, J., Liu, Z., & Sun, M. (2024). Scaling Efficient Masked Autoencoder Learning on Large Remote Sensing Dataset. *ArXiv, abs/2406.11933*. https://doi.org/10.48550/arxiv.2406.11933

Wang, X., Zhang, Y., Zhang, Z., Luo, Q., & Yang, J. (2022). GSC-MIM: Global semantic integrated self-distilled complementary masked image model for remote sensing images scene classification. https://doi.org/10.3389/fevo.2022.1083801

Xie, Y., Wang, Z., Chen, W., Li, Z., Jia, X., Li, Y., Wang, R., Chai, K., Li, R., & Skakun, S. (2024). When are Foundation Models Effective? Understanding the Suitability for Pixel-Level Classification Using Multispectral Imagery. *ArXiv, abs/2404.11797*. https://doi.org/10.48550/arxiv.2404.11797

Xu, X., Feng, Z., Yang, S., Li, M., & Yao, G. (2025). GA-MAE: Gradient-Guided Activation-Aware Masked Autoencoder for Remote Sensing Image Classification. *IEEE Transactions on Geoscience and Remote Sensing, 63*, 1-16. https://doi.org/10.1109/tgrs.2025.3637094

Zhang, M., Liu, Q., & Wang, Y. (2023). CtxMIM: Context-Enhanced Masked Image Modeling for Remote Sensing Image Understanding. *ACM Transactions on Multimedia Computing, Communications and Applications, 21*, 1 - 22. https://doi.org/10.1145/3769084

Zhang, T., Zhuang, Y., Chen, H., Chen, L., Wang, G., Gao, P., & Dong, H. (2023). Object-Centric Masked Image Modeling-Based Self-Supervised Pretraining for Remote Sensing Object Detection. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 16*, 5013-5025. https://doi.org/10.1109/jstars.2023.3277588

Zhang, Y., Zhao, Y., Dong, Y., & Du, B. (2023). Self-Supervised Pretraining via Multimodality Images With Transformer for Change Detection. *IEEE Transactions on Geoscience and Remote Sensing, 61*, 1-11. https://doi.org/10.1109/tgrs.2023.3271024

Zhang, X., Li, B., Zhou, C., Yu, W., & Zhang, L. (2026). SIGMAE: A Spectral-Index-Guided Foundation Model for Multispectral Remote Sensing. *ArXiv, abs/2603.07463*. https://doi.org/10.48550/arxiv.2603.07463